

# Lotus Leaf Surface Sprint

Engineer a self-cleaning, water-shedding surface

**At a glance:** Build a surface that sheds water and dirt the way a lotus leaf does. Plan about 80 minutes for the core block; 4 teams of 4 students.

## LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: hydrophobicity, surface roughness, biomimicry, materials engineering (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

## MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- wax paper: about 2 sheets per team per run (1 roll for the camp)
- cardstock: about 3 sheets per team per run (16 total; from the shared cardstock stock)
- sandpaper strips: medium grit, about 2 per team plus spares (12 total)
- masking tape: to fix surfaces to the ramp (shared roll)
- pipettes: 3 mL graduated, 1 to 2 per team (one pack of 100 covers the camp; reused)
- water: tap, a few mL per droplet (fill jug plus pour-off bucket)
- pepper or cocoa: washable dirt proxy, a pinch per run (1 small shaker each; source locally, not on the buy line, missing from the kit list)
- candle or paraffin wax: rubbed on cold to coat a surface (1 to 2 candles, shared; source locally, not on the buy line)
- ramps: 1 per team at a fixed standard tilt (on-hand rigid boards; missing from the kit list)

## PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

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- 01 Set up one station per team with the materials above, sorted and ready, including a ramp built from an on-hand rigid board.
- 02 Set a fixed standard tilt for every ramp (a block or book under one edge, or a taped angle mark) so all teams run at the same slope, fill a water jug and stage a pour-off bucket, and put out the scoring sheet before testing begins.
- 03 Put out a containment tray at each station and have gloves available for students who do not want pepper, cocoa, or wax on their hands. No goggles are needed: this is dropping water onto a tilted surface over a tray, not a chemical or splash hazard.
- 04 Load the activity slides and ready the projected timer.

## MINUTE-BY-MINUTE FACILITATION

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### Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Build a surface that sheds water and dirt the way a lotus leaf does." Take a quick prediction by show of hands.

### Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

### Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

### Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

### Phase 5 Test and record

Teams set the ramp to the standard tilt, release a measured droplet from the top, time the runoff over the tray, and score the residue left behind; they log this baseline before changing anything.

### Phase 6 Redesign

Each team changes exactly one variable (texture, amount of wax, tilt, or surface material), predicts the effect, and re-runs at the same tilt with the same droplet and dirt amount to compare against baseline.

### Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, then reset the station: bin the used paper surfaces and dirt-laden wax paper, pour off standing water, wipe the ramp and tray clear of grit and dirt, dry the table, and wash hands (remove gloves first if worn).

## TROUBLESHOOTING

**Problem:** A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

**Problem:** The droplet soaks in or will not roll on the first run. **Fix:** Confirm the surface has both features (a rough texture and a wax coating, not one alone), check the ramp is at the standard tilt and not too shallow, use a smaller droplet, and log the baseline anyway.

**Problem:** Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

**Problem:** A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

## DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to hydrophobic micro-texture.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

## DEBRIEF QUESTIONS (PICK 2 TO 3)

- Why does a rough, waxy surface shed water better than a smooth one?
- What carried the dirt away?
- Where would a self-cleaning surface be useful?

## SCORING RUBRIC (100 POINTS)

| SCORING CRITERION          | POINTS     |
|----------------------------|------------|
| Fast clean runoff          | 35         |
| Least residue              | 25         |
| Surface design explanation | 20         |
| Redesign gain              | 10         |
| Craftsmanship              | 10         |
| <b>Total</b>               | <b>100</b> |

## CLEANUP AND SOURCES

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Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

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**SOURCES** AskNature, Surface allows self-cleaning (lotus leaf); Koch and Barthlott, superhydrophobicity of the lotus leaf (contact angle about 163 degrees, hierarchical micro-papillae plus nano wax tubules).